

MedSphere – Unified Hospital Information Management System

MCA Semester 3 Mini Project Synopsis

Introduction

MedSphere is a Semester 3 MCA mini project providing a centralized web-based system for core hospital information and workflows. The current version covers patients, departments, doctors, appointments, consultations, prescriptions and medical history.

Problem Statement

Connected hospital workflows require consistent handling of patient, doctor, appointment and clinical information. MedSphere provides an academic centralized system for these core workflows through a role-based database-backed web application.

Objectives

- Centralize core hospital information.
- Provide role-based access for ADMIN, RECEPTIONIST and DOCTOR.
- Manage patients, departments and doctors.
- Schedule and manage appointments.
- Record consultations and prescriptions.
- Maintain patient medical history.
- Apply validation and business rules.
- Demonstrate secure authentication and authorization.

Technology Stack

Java 21; Spring Boot 3.5.5; Spring MVC; Spring Security; Spring Data JPA; Hibernate; Thymeleaf; Bootstrap 5.3.3; JavaScript; MariaDB/MySQL; Maven Wrapper; XAMPP.

Architecture

Browser/UI → Controller → Service → Repository → JPA/Hibernate → MariaDB/MySQL.

SDLC

Planning → Requirement Analysis → System Design → Implementation → Testing → Deployment/Operation → Maintenance/Future Enhancement.

Expected Outcome

A working academic hospital information management application demonstrating secure role-based access and end-to-end database-backed workflows.

Conclusion

MedSphere demonstrates how Java and Spring Boot technologies can be combined to implement a structured hospital information management application within Semester 3 scope.